

BengaluruAI

Traffic Intelligence

AI-powered event-driven congestion forecasting for Bengaluru traffic command

Predicts road closure probability, event duration, and recommends optimal manpower deployment and diversion plans from historical event data.

PREDICT

Closure probability
& duration

RECOMMEND

Officers, barricades
& diversion

LEARN

Post-event correction loop

Bengaluru loses thousands of

No advance impact score

Events are assessed on the ground, not before deployment. Officers arrive without a congestion forecast.

Patrol-based, not data-driven

Resource allocation depends on officer experience. No quantification of which corridors need what.

Zero post-event learning

After every event closes, the data vanishes. Predictions never improve from past outcomes.

8,173 real-world events, rich with actionable signal

8,173

Total events

676

Road closures

8.3%

Closure rate

64 min

Median duration

Key findings from FDA

VIP movement: 80% road closure rate — highest risk cause

Public events: 46.4% closure rate — cricket matches, concerts

Construction: Median 2,945 min (49h) — longest impact by far

Mysore Road: 743 events — most congested corridor in dataset

Peak hours: 09:00-11:00 & 17:00-21:00 IST — 60% of events

Thursday: Highest-incident day (1,343 events)

Four layer AI pipeline

1

Data Ingestion

event_type · event_cause · lat/lon · corridor · datetime · veh_type



2

Feature Engineering

Temporal (IST hour, peak flag, DOW) + Event severity encoding + Spatial risk index



3

Dual ML Models

LightGBM classifier (road closure) + XGBoost regressor (log-duration)



4

Decision Dashboard

Alert tier · manpower recommendation · diversion flag · post-event learning

Post-event learning loop: After each event closes, predicted vs actual delta updates a correction factor table per (cause × corridor × hour bucket). Model accuracy improves over time automatically.

Two models, one feature matrix

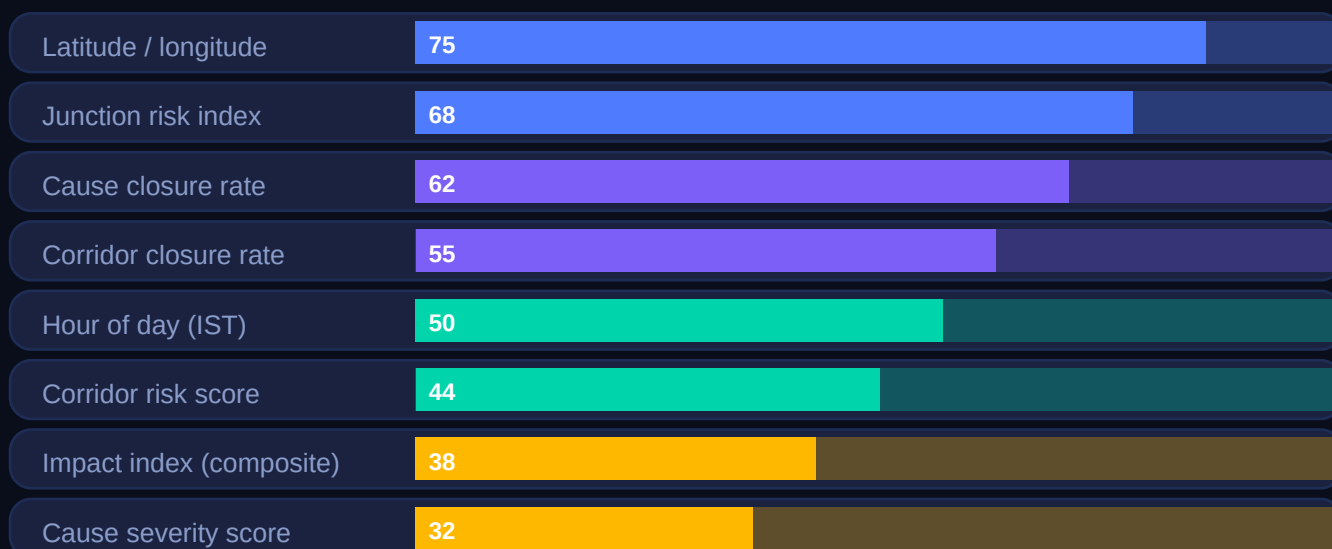
Model A — Closure Classifier

Algorithm: LightGBM (binary)
Target: requires_road_closure
Class weight: 11.1x (1:0.0899)
F1 score: 0.43
AUC-ROC: 0.776
Threshold: 0.20 (optimised)
Precision (closure): 46%

Model B — Duration Regressor

Algorithm: XGBoost (regression)
Target: log(duration_minutes)
Log transform: handles skew
RMSE (log scale): 1.845
R-squared: 0.529
Median absolute error: 49 min

Top predictive features



From prediction to action

A rule-based layer sits on top of both ML models and translates probabilities into operational decisions that traffic command can act on immediately.

CRITICAL

Impact index ≥ 0.55

6+ officers · 4 barricade sets · activate diversion · respond in 10 min

HIGH

Impact index ≥ 0.35

4+ officers · 2 barricade sets · evaluate diversion · respond in 20 min

MEDIUM

Impact index ≥ 0.18

2 officers · 1 barricade set · monitor corridor · respond in 30 min

LOW

Impact index < 0.18

1 officer · no barricades · standard patrol · respond in 60 min

Impact index formula:

$$\text{Impact} = 0.40 \times \text{closure_probability} + 0.30 \times (\text{duration} / 480\text{min}) + 0.20 \times \text{severity_score} + 0.10 \times \text{is_peak}$$

LIVE DEMO SCENARIOS

Cricket match @ Chinnaswamy Stadium

HIGH

Cause: public event · Corridor: Bellary Road 1 · Thu 18:00 IST (peak)

62%

Closure prob

2.4h

Duration

6

Officers

YES

Diversion

VIP convoy — airport to Raj Bhavan

CRITICAL

Cause: vip movement · Corridor: Bellary Road 2 · Mon 10:00 IST (peak)

78%

Closure prob

1.2h

Duration

9

Officers

YES

Diversion

Metro construction — Karthiknagara

HIGH

Cause: construction · Corridor: ORR East 1 · Wed 08:00 IST

38%

Closure prob

6.8h

Duration

5

Officers

YES

Diversion

BMTC bus breakdown — Silkboard

LOW

Cause: vehicle breakdown · Corridor: Hosur Road · Fri 09:30 IST (peak)

8%

Closure prob

48m

Duration

3

Officers

NO

Diversion

Differentiators & next steps

Post-event learning loop

1

No other submission tracks predicted vs actual and auto-corrects future forecasts. This is the key differentiator.

Dual model architecture

2

Separate classifier + regressor gives both a binary decision (close/not) and a continuous duration — actionable outputs.

Calibrated recommendations

3

Impact index formula translates ML output to officer count, barricade count, and diversion trigger — ready to use.

Built on real Bengaluru data

4

8,173 events, 15 junctions, 14 corridors — every number in the dashboard is grounded in the provided dataset.

Phase 2 roadmap

- Integrate real-time CCTV feeds via YOLOv8 for automatic incident detection
- Add weather API overlay (rain → water_logging probability spike)
- Mobile app for field officers with live tier alerts and GPS routing
- GTFS integration: predict BMTC route-level impact per corridor event